

Department of Electrical & Electronic Engineering

BRAC University

Semester: Spring 25



EEE101L

ELECTRICAL CIRCUITS I LABORATORY

Section: 05

Experiment Name: Verification of KVL and KCL

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Group Numbers: 05

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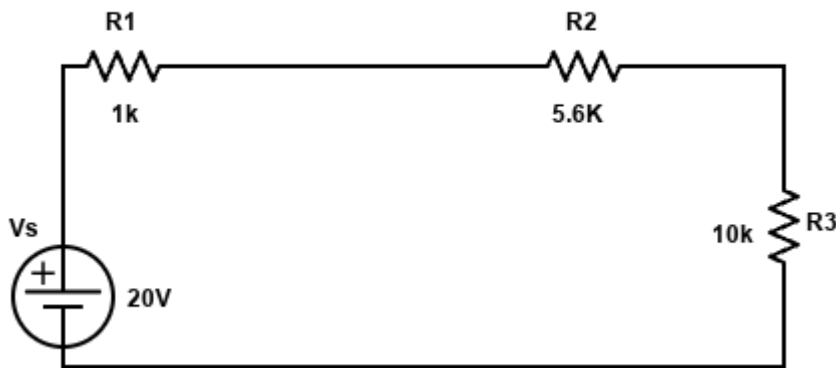
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Part-A: Verification of KVL

Objective: This experiment is intended to verify Kirchoff's voltage law (KVL) with the help of a series circuit.

Theoretical Background: KVL states that around any closed circuit the algebraic sum of the voltage rises equals the algebraic sum of the voltage drops.

Compilation of Result and Calculations:



V_s (V)	V_1 (V)	V_2 (V)	V_3 (V)	$V_1 + V_2 + V_3$ (V)
20	1.2	6.6	12.1	19.90

$$\therefore \text{Error} = \{ (20 - 19.90) / 20 \} \times 100 \% = 0.5 \%$$

Discussion:

Resistors have manufacturing tolerances. This means that their actual resistance may not match the nominal values exactly which leads to variations in expected voltage drops. Since the error is only 0.5%, it is quite small and within an acceptable range for practical measurements.

Answer to Lab Report Questions:

Answer 1 :

Voltmeter: Connect in parallel with high resistance.

Ammeter: Connect in series with low resistance.

Answer 2 :

$R_1 = 1 \text{ k}\Omega$; $R_2 = 5.6 \text{ k}\Omega$; $R_3 = 10 \text{ k}\Omega$; $V_s = 20\text{V}$; $V_1 = 1.2 \text{ V}$; $V_2 = 6.6 \text{ V}$;

$V_3 = 12.1 \text{ V}$

We know that,

Power, $p = (V)^2 / R$;

$$\therefore p_1 = (V_1)^2 / R_1 = (1.2)^2 / 1 \times (10)^3 = 1.44 \times (10)^{-3} \text{ W}$$

$$\therefore p_2 = (V_2)^2 / R_2 = (6.6)^2 / 5.6 \times (10)^3 = 7.78 \times (10)^{-3} \text{ W}$$

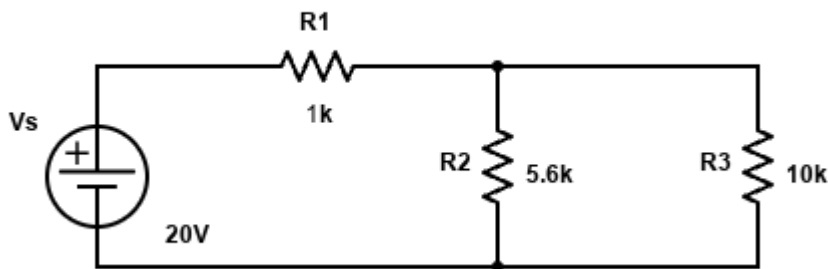
$$\therefore p_3 = (V_3)^2 / R_3 = (12.1)^2 / 10 \times (10)^3 = 14.64 \times (10)^{-3} \text{ W}$$

Part-B: Verification of KCL

Objective: This experiment is intended to verify Kirchoff's current law (KCL) with the help of a series parallel circuit.

Theoretical Background: KCL states that the algebraic sum of the currents entering any node equals the sum of the currents leaving the node.

Compilation of Result and Calculations:



R1 (k Ω)	R2 (k Ω)	R1 (k Ω)	V1 (V)	V2 (V)	I1 = V1 / R1 (mA)	I2 = V2 / R2 (mA)	I3 = V2 / R3 (mA)	I2 + I3 (mA)
1.04	5.53	10.11	4.5	15.5	4.33	2.80	1.53	4.33

$$\therefore \text{Error} = \{ I_1 - (I_2 + I_3) / I_1 \} \times 100 \% = \{ 4.33 - (4.33) / 4.33 \} \times 100 \% = 0 \%$$

Discussion:

Since the error is 0 % , there is no discrepancy.

Answer to Lab Report Questions:

Answer 1 :

$$R1 = 1.04 \text{ k}\Omega ; R2 = 5.5 \text{ k}\Omega ; R3 = 10.11 \text{ k}\Omega ; V1 = 4.5 \text{ V} ; V2 = 15.11 \text{ V} ;$$

$$V_s = 20 \text{ V}$$

R2 and R3 are in parallel,

$$\therefore (1 / R_p) = (1 / R_2) + (1 / R_3) \text{ unit}$$

$$\text{or, } (1 / R_p) = (1 / 5.5) + (1 / 10.11) \text{ k}\Omega$$

$$\text{or, } (1 / R_p) = 3122 / 11121 \text{ k}\Omega$$

$$\text{or, } R_p = 11121 / 3122 \text{ k}\Omega$$

$$\text{or, } R_p = 3.562 \text{ k}\Omega ;$$

Rp and R1 are in series,

$$\therefore R_s = R_p + R_1 \text{ unit}$$

$$\text{or, } R_s = (3.562 + 1.04) \text{ k}\Omega$$

$$\text{or, } R_s = 4.602 \text{ k}\Omega$$

$$\text{or, } R_s = 4.602 \times 10^3 ;$$

Power supplied by the sources,

$$P = \{ (V_s)^2 / R_s \} \text{ unit}$$

$$\text{or, } P = (20)^2 / \{ 4.602 \times 10^3 \} \text{ W} ;$$

$$\text{or, } P = 0.869 \text{ W} ;$$

Power absorbed by each of the resistor is,

$$P_1 = \{ (V_1)^2 / R_1 \} \text{ unit}$$

$$\text{or, } P_1 = (4.5)^2 / \{ 1.04 \times 10^3 \} \text{ W}$$

$$\text{or, } P_1 = 0.0195 \text{ W} ;$$

$$P_2 = \{ (V_2)^2 / R_2 \} \text{ unit}$$

$$\text{or, } P_2 = (15.5)^2 / \{ 5.5 \times 10^3 \} \text{ W}$$

$$\text{or, } P_2 = 0.0457 \text{ W} ;$$

$$P_3 = \{ (V_2)^2 / R_3 \} \text{ unit}$$

$$\text{or, } P_3 = (15.5)^2 / \{ 10.11 \times 10^3 \} \text{ W}$$

or, $P_3 = 0.0238 \text{ W}$;

Answer 2 :

Power supplied by the sources, $P = 0.869 \text{ W}$;

Total power absorbed by the resistor, $P' = (P_1 + P_2 + P_3) \text{ unit}$;

or, $P' = (0.0195 + 0.0457 + 0.0238) \text{ W}$

or, $P' = 0.089 \text{ W}$

$\therefore P \neq P'$

So, the supplied power is not equal to total power absorbed.

Appendix:

Electrical Circuits I Laboratory

EEE 101L

Department of Electrical & Electronic
Engineering (EEE) Brac University

Experiment No. 1

Verification of KVL and KCL

Part-A: Verification of KVL

1. **Objective:** This experiment is intended to verify Kirchhoff's voltage law (KVL) with the help of series circuits.
2. **Theoretical Background:** KVL states that around any closed circuit the algebraic sum of the voltage rises equals the algebraic sum of the voltage drops.
3. **Equipment:**
 - One multimeter
 - Three Resistors
 - One DC power supply

1. Cover Page
2. Objective
3. Theoretical Background
4. Compilation of result and calculations (with ckt diagram and per verification of result).
5. Answer to lab report questions.
6. Discussion (must explain reasons for discrepancy).
7. Conclusion
8. Appendix (Signed Datasheet).



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$$-V_S + V_1 + V_2 + V_3 = 0$$

$$V_1 + V_2 + V_3 = V_S$$

1

4. Circuit Diagram:

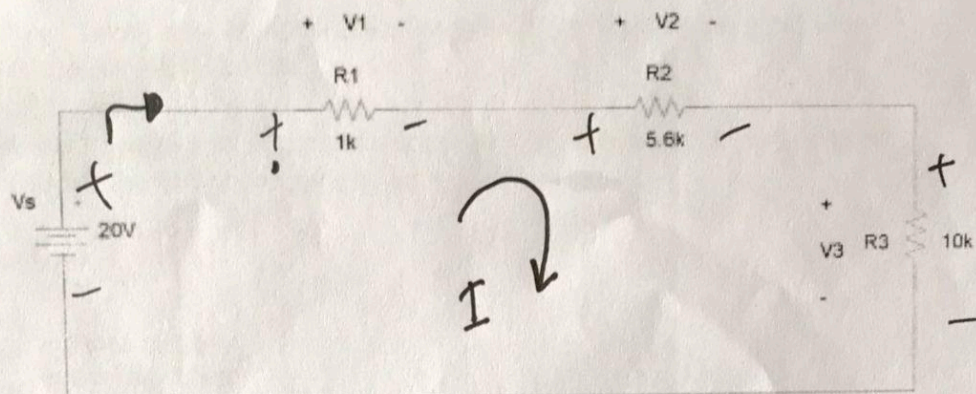


Fig. 1: Series circuit to demonstrate KVL

5. Procedure:

- Connect the resistors R_1 , R_2 and R_3 in series to a DC power supply as shown in Fig. 1.
- Take readings of V_1 , V_2 , V_3 , V_S using a multimeter.
- Verify KVL as $V_S = V_1 + V_2 + V_3$.

6. Data Table:

Verification of KVL:

V_S (V)	V_1 (V)	V_2 (V)	V_3 (V)	$V_1 + V_2 + V_3$ (V)
20	1.2	6.6	12.1	19.9

Faculty Signature and Date

7. Lab report directions:

- State the rules of connecting voltmeter and ammeter in the circuit.
- How do you calculate the power dissipation of a resistor?

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Part-B: Verification of KCL

1. Objective: This experiment is intended to verify Kirchhoff's current law (KCL) with the help of a series parallel circuit.

2. Theoretical Background: KCL states that the algebraic sum of the currents entering any node equals the sum of the currents leaving the node.

3. Equipment:

- Three resistors
- One multimeter
- One DC supply

4. Circuit Diagram:

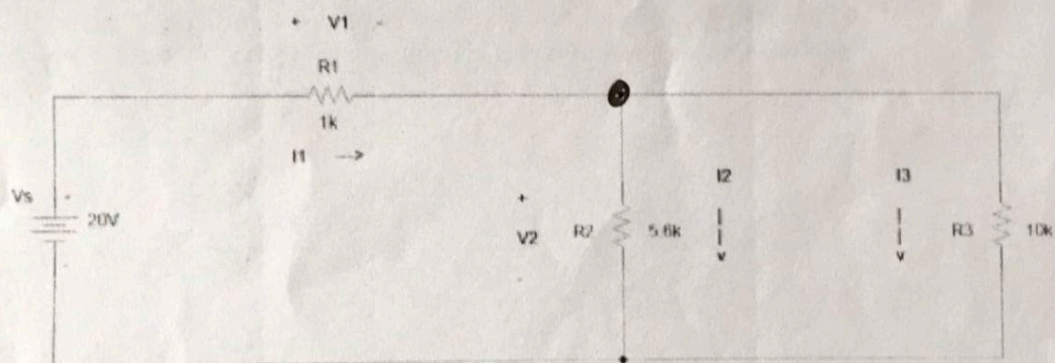


Fig. 2: Verification of KCL

5. Procedure:

- i. Connect the resistors in series-parallel across the power supply as shown in Fig. 2.
- ii. Measure the voltage drop across the resistors.
- iii. Calculate the currents I_1 , I_2 and I_3 by using $I_1 = V_1/R_1$, $I_2 = V_2/R_2$ and $I_3 = V_3/R_3$ as given above and prove that $I_1 = I_2 + I_3$.

V/R

3

6. Data Table:

Verification of KCL:

R_1 ($k\Omega$)	R_2 ($k\Omega$)	R_3 ($k\Omega$)	V_1 (V)	V_2 (V)	$I_1 = V_1/R_1$ (mA)	$I_2 = V_2/R_2$ (mA)	$I_3 = V_2/R_3$ (mA)	$I_2 + I_3$ (mA)
1.04	9.50	10.11	4.5	15.9	4.33	2.80	1.50	4.33

Faculty Signature and Date

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7. Lab report directions:

1. Calculate the total power supplied by the source and power absorbed by each of the resistor. 2. Is the supplied power equal to total power absorbed?